

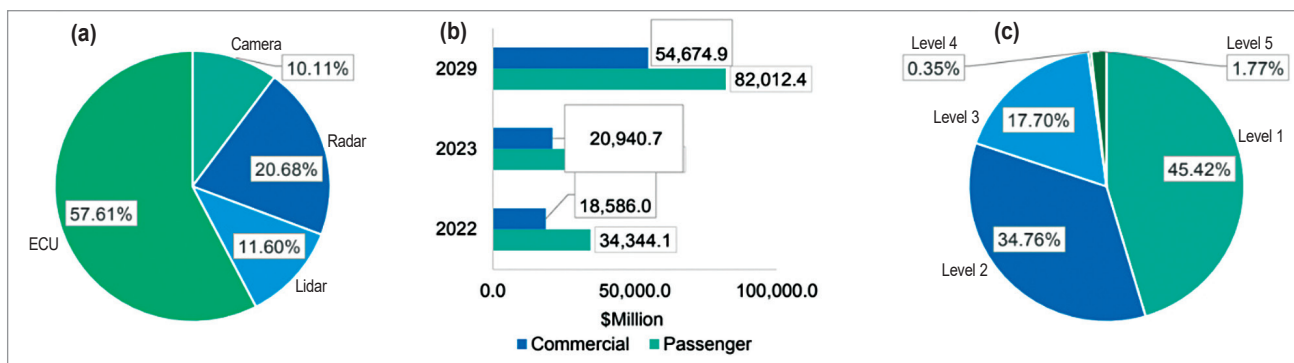


Fig. 5: (a) Global advanced sensors (by component type), value share, 2022 (% of global revenue), (b) Global advanced sensors (by vehicle type), and (c) Global advanced sensors (by level of autonomy), value share, 2022 (% of global revenue)

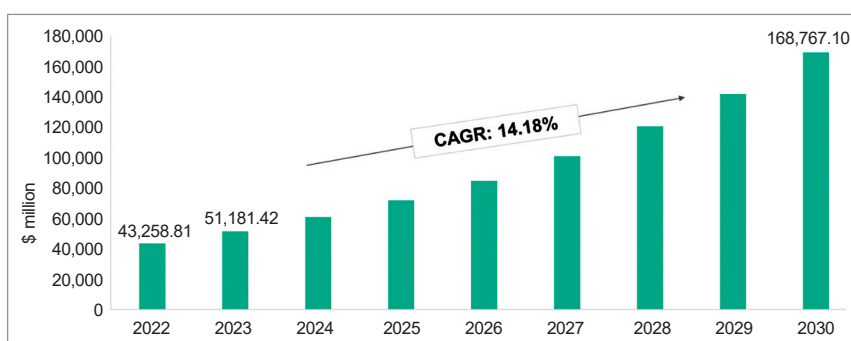


Fig. 6: Global power electronics market scenario—present and projected

sensors more than doubling. This surge can be attributed to the escalating adoption of level two, level three, and higher-level autonomous vehicles, which increasingly integrate advanced sensors into their architecture. Regarding segmental shares, camera sensors dominate the landscape, especially as they find widespread use in India's vehicles, serving various functions, from essential parking assistance to advanced driver assistance systems (ADAS).

Furthermore, passenger vehicles currently lead in adopting advanced sensors, while commercial vehicles have lagged due to cost considerations. However, this trend is expected to change as companies like Volvo and others increasingly incorporate these sensors into their commercial vehicle offerings. This shift is projected to boost their market share significantly by 2029.

These three categories stand out as the primary drivers, contributing to over 50% of the revenue generated by sensor manufacturers worldwide. When examining the segmentation by vehicle type, it becomes evident that passenger vehicles, owing to their sheer numbers, have consistently held the leading position and are anticipated to maintain this dominance well into 2030. Additionally, when delving into the application type breakdown, the pie chart illustrates that powertrain-based sensors within the conventional sensor category command a substantial share, occupying nearly 40% of the overall market. Regarding autonomy levels, as of 2022, level one autonomy holds the largest share, accounting for approximately 42%, with higher levels of autonomy gradually gaining ground.

Turning our attention to power electronics, several key drivers,

restraints, and opportunities merit discussion, focusing on points we have not yet covered. Among the observed constraints is the rising cost of raw materials. While semiconductor costs have generally decreased over the past decade, our discussions with clients have revealed that the increased raw material costs primarily stem from logistical challenges, partnerships, and manufacturing hub decisions. Many companies actively address these concerns by enhancing their global distribution networks and securing robust supplier relationships.

Regarding market dynamics, the power electronics sector has experienced substantial growth, driven mainly by the combined demand from electric vehicles, connected vehicles, and autonomous vehicles. Regardless of vehicle type, power electronics play a pivotal role. This increased adoption across the automotive spectrum is forecasted to sustain a healthy growth rate of 14.18% over the next decade, culminating in 2030.

The Asia Pacific region, particularly China, holds a dominant position in the automotive electronics market due to its high vehicle production volume. Europe and North America also have significant market shares, thanks to their autonomous vehicle manufacturing and supportive regulations. In